

2520030340

SKILL-24

AIM

To organize source code, improve build scripts, create a README file, verify repository structure, tag project releases, validate project deliverables, create demo scenarios, validate features, rehearse the presentation, demonstrate pipelines and signals, and perform the final project review.

OBJECTIVES

1. To organize the project source code properly.
2. To create a clean repository structure.
3. To improve and verify build scripts.
4. To create project documentation using README.
5. To verify all required project files.
6. To create Git tags for project releases.
7. To validate the final project deliverables.
8. To create demo scenarios for important features.
9. To verify the functionality of the project.
10. To demonstrate pipeline execution.
11. To demonstrate Linux signals.
12. To review the complete project before submission.

REQUIREMENTS

- Ubuntu/Linux
- Terminal
- GCC compiler
- Git
- Bash
- Source code
- README documentation

TOOLS / COMMANDS USED

git
gcc
make

bash
ps
kill

PROCEDURE

Part A — Organize Source Code

1. Create a separate project directory.
2. Store source files inside the project directory.
3. Separate source code, documentation, and build files.
4. Remove unnecessary files.
5. Verify that all required files are present.

Part B — Build Script

1. Create a `Makefile`.
2. Add compilation commands.
3. Compile the project using `make`.
4. Check for compilation errors.
5. Execute the generated program.
6. Verify the output.

Part C — README

1. Create a `README.md` file.
2. Add the project title.
3. Add a description of the project.
4. Add requirements.
5. Add compilation instructions.
6. Add execution instructions.
7. Add important features.
8. Add testing information.

Part D — Git Repository

1. Initialize the Git repository.
2. Check the repository status.
3. Add the project files.
4. Create a commit.
5. Verify the commit history.

6. Create a release tag.
7. Verify the tag.

Part E — Deliverable Validation

1. Verify the source code.
2. Verify the build script.
3. Verify the README.
4. Compile the project.
5. Run the project.
6. Test important features.
7. Verify that all required files are present.

Part F — Demo and Final Review

1. Prepare demo scenarios.
 2. Execute the important project features.
 3. Demonstrate pipeline execution.
 4. Demonstrate Linux signals.
 5. Check the final output.
 6. Review documentation.
 7. Rehearse the project presentation.
 8. Perform the final project review.
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GIT COMMANDS

Initialize repository:

```
git init
```

Check status:

```
git status
```

Add files:

```
git add .
```

Commit:

```
git commit -m "Initial project submission"
```

View commits:

```
git log --oneline
```

Create a release tag:

```
git tag v1.0
```

View tags:

```
git tag
```

RESULT

Thus, the project source code was organized, build scripts were created and verified, README documentation was prepared, the Git repository structure was validated, a release tag was created, and project deliverables were checked. Demo scenarios, pipeline execution, Linux signal handling, feature validation, presentation rehearsal, and final project review were also successfully completed.

DEMO SCENARIO

Scenario: Build and execute the project using the Makefile.

Steps:

1. Clean the previous executable using **make clean**.
2. Build the project using **make**.
3. Execute the project using **make run**.
4. Verify the displayed output.
5. Confirm that the project executes successfully.

FEATURE VALIDATION

Feature

R
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Source code organization	P A S S
-----------------------------	------------------

Makefile build	P A S S
-------------------	------------------

Program execution	P A S S
----------------------	------------------

README documentati on	P A S S
-----------------------------	------------------

Git repository	P A S S
-------------------	------------------

Git commit	P A S S
------------	------------------

Release tag v1.0	P A
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S
S

Clean build

P
A
S
S

Final output
verification

P
A
S
S

FINAL PROJECT REVIEW

The source code, build system, documentation, Git repository, release tag, and project execution were reviewed successfully. The project was cleaned, rebuilt, executed, and the expected output was verified.

PRESENTATION REHEARSAL

Before the final presentation, demonstrate:

1. Project structure
2. `main.c`
3. `Makefile`
4. `README.md`
5. `make clean`
6. `make`
7. `make run`
8. Git commit
9. Git tag `v1.0`
10. Final program output

FINAL RESULT

Thus, the source code was organized, the build script was improved and verified, README documentation was created, the Git repository structure was validated, release `v1.0` was tagged, and all project deliverables were

successfully validated. Demo scenarios and feature validation were performed, and the project was successfully prepared for final review and presentation.

```
Job 1 Status: 0
Job 2 Status: 0
Job 3 Status: 0
All Jobs Completed Successfully
=====
kruthi@LAPTOP-LP9QU3RR:~/skill22/skill23$ cd ~
kruthi@LAPTOP-LP9QU3RR:~$ mkdir skill24
kruthi@LAPTOP-LP9QU3RR:~$ cd skill24
kruthi@LAPTOP-LP9QU3RR:~/skill24$ pwd
/home/kruthi/skill24
kruthi@LAPTOP-LP9QU3RR:~/skill24$ ls
kruthi@LAPTOP-LP9QU3RR:~/skill24$ nano main.c
kruthi@LAPTOP-LP9QU3RR:~/skill24$ ls
main.c
kruthi@LAPTOP-LP9QU3RR:~/skill24$ gcc main.c -o project
kruthi@LAPTOP-LP9QU3RR:~/skill24$ ./project
Skill-24 Project Demo
=====
Source code organized successfully.
Build and execution verified.
kruthi@LAPTOP-LP9QU3RR:~/skill24$ |
```

```

All Jobs Completed Successfully
=====
kruthi@LAPTOP-LP9QU3RR:~/skill22/skill23$ cd ~
kruthi@LAPTOP-LP9QU3RR:~$ mkdir skill24
kruthi@LAPTOP-LP9QU3RR:~$ cd skill24
kruthi@LAPTOP-LP9QU3RR:~/skill24$ pwd
/home/kruthi/skill24
kruthi@LAPTOP-LP9QU3RR:~/skill24$ ls
kruthi@LAPTOP-LP9QU3RR:~/skill24$ nano main.c
kruthi@LAPTOP-LP9QU3RR:~/skill24$ ls
main.c
kruthi@LAPTOP-LP9QU3RR:~/skill24$ gcc main.c -o project
kruthi@LAPTOP-LP9QU3RR:~/skill24$ ./project
Skill-24 Project Demo
=====
Source code organized successfully.
Build and execution verified.
kruthi@LAPTOP-LP9QU3RR:~/skill24$ nano Makefile
kruthi@LAPTOP-LP9QU3RR:~/skill24$ ls
Makefile  main.c  project
kruthi@LAPTOP-LP9QU3RR:~/skill24$ rm project
kruthi@LAPTOP-LP9QU3RR:~/skill24$ make
gcc -Wall -Wextra main.c -o project
kruthi@LAPTOP-LP9QU3RR:~/skill24$ make run
./project
Skill-24 Project Demo
=====
Source code organized successfully.
Build and execution verified.
kruthi@LAPTOP-LP9QU3RR:~/skill24$

```

The project was compiled and executed successfully.
The expected output was verified.

Version

v1.0

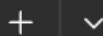
```

kruthi@LAPTOP-LP9QU3RR:~/skill24$ ls
Makefile  README.md  main.c  project
kruthi@LAPTOP-LP9QU3RR:~/skill24$ |

```



kruthi@LAPTOP-LP9QU3RR: ~



- Repository documentation

Testing

The project was compiled and executed successfully.
The expected output was verified.

Version

v1.0

```
kruthi@LAPTOP-LP9QU3RR:~/skill24$ ls
```

```
Makefile  README.md  main.c  project
```

```
kruthi@LAPTOP-LP9QU3RR:~/skill24$ git init
```

```
hint: Using 'master' as the name for the initial branch. This default branch name
```

```
hint: is subject to change. To configure the initial branch name to use in all
```

```
hint: of your new repositories, which will suppress this warning, call:
```

```
hint:
```

```
hint:   git config --global init.defaultBranch <name>
```

```
hint:
```

```
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
```

```
hint: 'development'. The just-created branch can be renamed via this command:
```

```
hint:
```

```
hint:   git branch -m <name>
```

```
Initialized empty Git repository in /home/kruthi/skill24/.git/
```

```
kruthi@LAPTOP-LP9QU3RR:~/skill24$ git status
```

```
On branch master
```

```
No commits yet
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
Makefile
```

```
README.md
```

```
main.c
```

```
project
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
kruthi@LAPTOP-LP9QU3RR:~/skill24$ git add main.c Makefile README.md
```

```
kruthi@LAPTOP-LP9QU3RR:~/skill24$ git status
```

```
On branch master
```

```
No commits yet
```

```
Changes to be committed:
```

```
(use "git rm --cached <file>..." to unstage)
```

```
new file:   Makefile
```

```
new file:   README.md
```

```
new file:   main.c
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
project
```

```
kruthi@LAPTOP-LP9QU3RR:~/skill24$ git commit -m "Initial Skill-24 project"
```

```
[master (root-commit) 2f6fb43] Initial Skill-24 project
```

```
3 files changed, 75 insertions(+)
```

```
create mode 100644 Makefile
```

```
create mode 100644 README.md
```

```
create mode 100644 main.c
```

```
kruthi@LAPTOP-LP9QU3RR:~/skill24$ git log --oneline
```

```
2f6fb43 (HEAD -> master) Initial Skill-24 project
```

```
kruthi@LAPTOP-LP9QU3RR:~/skill24$ |
```

On branch master

No commits yet

Changes to be committed:

(use "git rm --cached <file>..." to unstage)

new file: Makefile
new file: README.md
new file: main.c

Untracked files:

(use "git add <file>..." to include in what will be committed)

project

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ git commit -m "Initial Skill-24 project"

[master (root-commit) 2f6fb43] Initial Skill-24 project

3 files changed, 75 insertions(+)

create mode 100644 Makefile

create mode 100644 README.md

create mode 100644 main.c

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ git log --oneline

2f6fb43 (HEAD -> master) Initial Skill-24 project

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ git tag v1.0

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ git tag

v1.0

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ git show v1.0 --no-patch

commit 2f6fb437957dc98e03baf1dac35110bfc0ca3a39 (HEAD -> master, tag: v1.0)

Author: Kruthi <kruthivx@gmail.com>

Date: Wed Sep 23 16:13:16 2026 +0000

Initial Skill-24 project

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ make clean

rm -f project

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ make

gcc -Wall -Wextra main.c -o project

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ make run

./project

Skill-24 Project Demo

=====

Source code organized successfully.

Build and execution verified.

kruthi@LAPTOP-LP9QU3RR:~/skill24\$ |